SEMINAR REPORT

A seminar Report on

AR in Education: Benefits & Challenges

Submitted partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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DEPARTMENT OF COMPUTER ENGINEERING

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TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2
3	Information/Chapters	3
4	Advantages and Disadvantages	8
5	Conclusion	9
6	Project Initiation:	10
	6.1) Description of problem statement.	10
	6.2) Scope and objectives.	10
	6.3) Stake holder	11
	6.4) Platform/ Equipment/ Resources identification.	12
7	Seminar Presentation	
	Bibliography	
	References	

ABSTRACT

Augmented Reality (AR) is a technology that overlays digital information—such as three-dimensional models, labels, animations, audio, and interactive objects—onto the real-world environment. In education, AR can help learners explore concepts that may be difficult to understand through textbooks or two-dimensional diagrams alone. For example, students can view a three-dimensional model of the human heart, examine geometric shapes from different angles, explore historical locations, or observe virtual science experiments using a smartphone, tablet, or AR headset.

This seminar report reviews the applications, benefits, and challenges of AR in education. It discusses how AR can support science, mathematics, language learning, geography, history, technical education, and skills training. Research reviews indicate that AR can improve engagement, motivation, interaction, spatial understanding, and learning outcomes when it is aligned with clear educational goals. However, its use also presents challenges, including device and development costs, limited connectivity, usability issues, teacher training needs, accessibility concerns, distractions, and cognitive overload.

LIST OF FIGURES

Fig. No.	Title	Page No.
1.1	Basic concept of Augmented Reality	4
3.1	Working process of an AR learning application	6

LIST OF TABLES

Table No.	Title	Page No.
2.1	Summary of selected literature	3
3.1	Applications of AR across subjects	5
6.1	Platforms, equipment and resources	14

CHAPTER 1

INTRODUCTION

Education has increasingly adopted digital tools to make teaching and learning more interactive, flexible, and learner centered. Traditional resources such as textbooks, diagrams, and classroom demonstrations remain important, but they may not always provide a clear view of complex, microscopic, three-dimensional, or distant objects. Augmented Reality (AR) can connect digital content with the physical world and allow students to explore information in context.

AR is commonly experienced through a mobile phone or tablet camera, which displays digital objects over a view of the real environment. Some systems use printed markers, while others use image recognition, surface detection, or location data. In an educational activity, students may point a device at a textbook illustration and see a labelled 3D model, rotate a virtual object, or follow step-by-step instructions.

Augmented Reality is a technology that adds computer-generated content to a view of the real world in real time. The added content may include text, images, 3D objects, sound, animation, or interactive simulations. Unlike Virtual Reality, which typically replaces the user's view of the real world with a computer-generated environment, AR keeps the real environment visible and supplements it with digital information.

1.3 Need for AR in Education

To make abstract concepts easier to visualize through 3D models and animations.

To increase student participation through exploration and hands-on interaction.

To support learning by doing, practice, and repeated observation.

To provide context-sensitive information for laboratories, workshops, museums, and field trips.

To supplement classroom explanations with visual and interactive learning materials.

1.4 Aim of the Report

The aim of this report is to study how Augmented Reality is used in education, identify its major benefits, and discuss the challenges that affect its adoption in schools, colleges, and other learning environments

1.5 Organization of the Report

The report first reviews relevant research, then explains AR concepts and applications. It next examines benefits and limitations before presenting the problem statement, scope, objectives, stakeholders, and resources for a possible educational AR initiative.

CHAPTER 2

LITERATURE REVIEW

This section summarizes selected research papers related to Augmented Reality in education. The summaries are based on the published articles and should be checked against the original papers before final submission.

1 Akçayır and Akçayır (2017)

Murat Akçayır, Gökçe Akçayır In "Advantages and Challenges Associated with Augmented Reality for Education: A Systematic Review of the Literature," Murat Akçayır and Gökçe Akçayır reviewed research on AR in educational settings. Their review reported that AR can support learning performance, while usability and difficulty of use were among the challenges reported by students. The paper is directly relevant because it considers both the benefits and limitations of educational AR.

2 Alzahrani (2020)

Nouf Matar Alzahrani Nouf Matar Alzahrani's "Augmented Reality: A Systematic Review of Its Benefits and Challenges in E-learning Contexts" reviewed 28 studies. The article describes benefits including learner engagement, motivation, attention, collaboration, knowledge retention, spatial ability, and access to information. It also identifies challenges such as cognitive overload, limited user experience, teacher resistance, cost, complex technology, and connectivity problems.

3 Garzón et al. (2020)

Juan Garzón, Kinshuk, Silvia Balderas, Jaime Gutiérrez, Juan Pavón "How Do Pedagogical Approaches Affect the Impact of Augmented Reality on Education? A Meta-analysis and Research Synthesis" examines the role of teaching approaches in AR-supported learning. It highlights that the educational impact of AR is linked to how technology is integrated with teaching strategies. This suggests that AR should support specific learning objectives rather than be added only for novelty.

4 Chang et al. (2022)

Hsin-Yi Chang, Theerapong Binali, Jyh-Chong Liang, Guo-Li Chiou, Kun-Hung Cheng, Silvia Wen-Yu Lee, Chin-Chung Tsai "Ten Years of Augmented Reality in Education: A Meta-analysis of (Quasi-) Experimental Studies to Investigate the Impact" reviews experimental research on AR in education. The paper investigates the relationship between AR use and learning outcomes and provides evidence that AR can be useful in educational interventions. Its findings also point to the importance of instructional design and study context when interpreting results.

5 Zhang et al. (2022)

Jiakai Zhang, Gege Li, Qinglin Huang, Qinna Feng, Heng Luo "Augmented Reality in K–12 Education: A Systematic Review and Meta-Analysis of the Literature from 2000 to 2020" reviewed 129 studies on AR in primary and secondary education. The authors examined application areas, technology features, instructional design, and research outcomes. The review reports that AR has been used in areas such as science, social science, and language learning, and that mobile devices are common delivery tools.

6 Additional review of AR applications

Ronald T. Azuma The wider body of AR education research explores mobile AR, marker-based learning, location-based learning, 3D visualization, and immersive experiences. Across these areas, findings suggest potential improvements in engagement and understanding, but outcomes vary according to the learner group, subject, duration, design, and quality of the AR content.

7 Sakr and Abdullah (2024)

Effie Lai-Chong Law, Matthias Heintz In "Virtual, Augmented Reality and Learning Analytics Impact on Learners, and Educators: A Systematic Review," Asmaa Sakr and Tariq Abdullah analyzed research on VR, AR, and learning analytics across educational settings. Their review found that these technologies can support learner motivation and attention, improve understanding and performance, and help knowledge retention. The paper also discusses how learning analytics can help educators understand learner interactions and improve learning experiences. This seminar is relevant because it considers both learner outcomes and support for teachers.

8 Composures (2024)

Asmaa Sakr, Tariq Abdullah In "Revealing the True Potential and Prospects of Augmented Reality in Education," Yiannis Composures reviews the educational potential and future prospects of AR. The paper discusses the growing interest in AR as a way to create more interactive and immersive learning experiences, while recognizing that successful adoption depends on suitable design and implementation. This study supports the view that AR can enrich teaching, but it should be selected according to learning goals, available resources, and the needs of students and teachers.

9 Additional reviews of AR applications

Yiannis Koumpouros The wider body of AR education research explores mobile AR, marker-based learning, location-based learning, 3D visualization, and immersive experiences. Across these areas, findings suggest potential improvements in engagement and understanding, but outcomes vary according to the learner group, subject, duration, design, and quality of the AR content.

Table No 2.1: Literature review

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Akçayır & Akçayır (2017) A review of augmented reality in science learning: advantages and challenges	Mobile/marker based AR, AR Learning applications	Review of studies on AR in Science education	To identify the benefits and challenges of using AR in science learning.	Reported challenges include usability issues, technical problems, and increased cognitive load.
2	Alzahrani (2020) Augmented Reality: A Review of Applications and Challenges in Education	AR applications on mobile devices and other Learning platforms	Review of AR Applications in education	To summaries educational uses of AR and the main challenges affecting implementation.	Findings depend on the studies reviewed; implementation can be affected by cost, device availability, and technical support.
3	Garzón et al (2020) Meta analysis of the impact of augmented reality on students' learning gains	AR-based learning tools and interactive 3D content	Meta analysis of experimental and quasi experimental studies	To estimate the effect of AR on students' learning outcomes.	Effects may vary by subject, age group, instructional design, and study quality.
4	Chang et al. (2022) Augmented reality in education: learning outcomes and related factors	AR learning environments, mobile AR and interactive visualization	Review/analysis of educational AR studies	To examine learning outcomes and factors associated with effective AR learning.	Results may not generalise across all subjects, learners, or AR designs.
5	Zhang et al. (2022) Augmented reality in education and Learning: a review	Mobile AR, head-mounted displays, 3D overlays	Review of Published AR-in Education research	To examine how AR is used in learning and identify trends and research needs.	AR hardware and software can be costly; privacy, accessibility, and usability require attention.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
6	Azuma (1997) A survey of augmented reality	Head-mounted displays, tracking, real time registration and 3D overlays	Technical Literature Survey (Foundation l AR paper)	To define AR and explain its key technical characteristics.	Not focused specifically, on classroom outcomes; It is an older foundational survey.
7	Law & Heintz (2021) applications for K–12 education: usability and user experience review	AR applications for school-age learners	Systematic review from usability and user experience perspectives	To review usability and user-experience issues in AR applications for K–12 education.	Evaluation methods and coverage of learners with special Educational needs are limited in the reviewed literature.
8	Sakr & Abdullah (2024) VR, AR and learning analytics: systematic Review	VR/AR learning environments and learning analytics	Systematic review of research involving learners and educators	To review how Immersive technologies and learning analytics relate to learning and teaching.	Combines multiple technology types; conclusions specifically about AR may be difficult to isolate.
9	Composures (2024) The potential and prospects of augmented reality in education	AR-enabled interactive and immersive learning experiences	Review/dis AR lesson potential and future prospects	To discuss Educational opportunities and future directions for AR.	Broad review; practical impact depends on implementation, teacher preparation, and available resources.

CHAPTER 3

INFORMATION ON SEMINAR

1.1 Marker-based AR

Marker-based AR uses a recognized image, symbol, QR-like marker, or printed target to trigger digital content. For example, scanning a marker in a textbook may display a 3D model of a planet or machine part.

1.2 Marker less AR

Marker less AR uses device sensors and computer vision to recognize surfaces or the surrounding environment. Students may place virtual objects on a desk or floor without needing a printed marker.

1.3 Location-based AR

Location-based AR uses location sensors such as GPS and a device compass to present information tied to a real place. It can support outdoor learning, geography, heritage trails, and educational field visits.

1.4 Projection-based and Superimposition AR

Projection-based AR projects light or images onto physical surfaces. Superimposition AR replaces or augments part of a real view with digital representation, which can be useful for demonstrations and training

1.Working Principle:

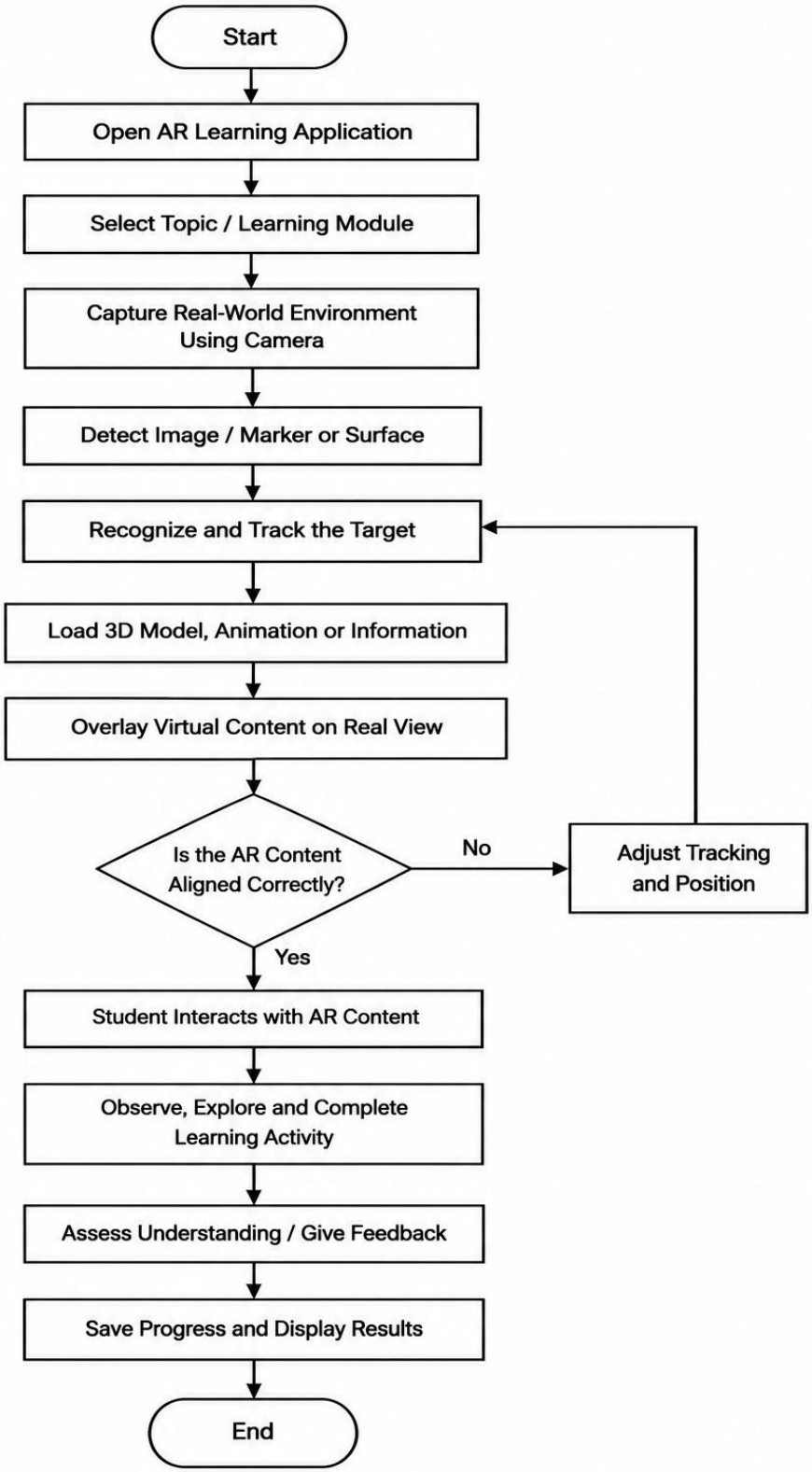


Fig. 3.1: Working of Principle on Augmented Reality in Education

The working of Augmented Reality (AR) in education, along with its applications, benefits, and challenges, consists of the following steps:

1. **Open AR Learning Application:** The student opens an AR application on a smartphone, tablet, or computer.

2. **Select Learning Topic:** The student selects a subject or learning module, such as Science, Mathematics, Biology, or Engineering.

3. **Capture Real-World Environment:** The device camera captures the surrounding environment or a printed image.

4. **Detect Image or Surface:** The AR application detects a marker, image, or surface to place virtual content.

5. **Recognize and Track the Target:** The application identifies and tracks the target to position virtual objects correctly.

6. **Load Virtual Content:** The application loads 3D models, animations, videos, or educational information related to the selected topic.

7. **Overlay Virtual Content:** Digital objects are displayed over the real-world view, creating an interactive learning experience.

8. **Interact with AR Content:** Students can rotate, zoom, explore, and interact with virtual objects to understand concepts more easily.

9. **Complete Learning Activities:** Students observe the virtual models, perform learning activities, and explore the topic.

10. **Assess Understanding:** The application may provide quizzes, exercises, or feedback to evaluate student understanding.

11. **Display Results:** The application displays the learning results and may save the student's progress.

12. **End:** The student completes the learning activity and gains a better understanding of the topic.

2.System Architecture and Main Components:

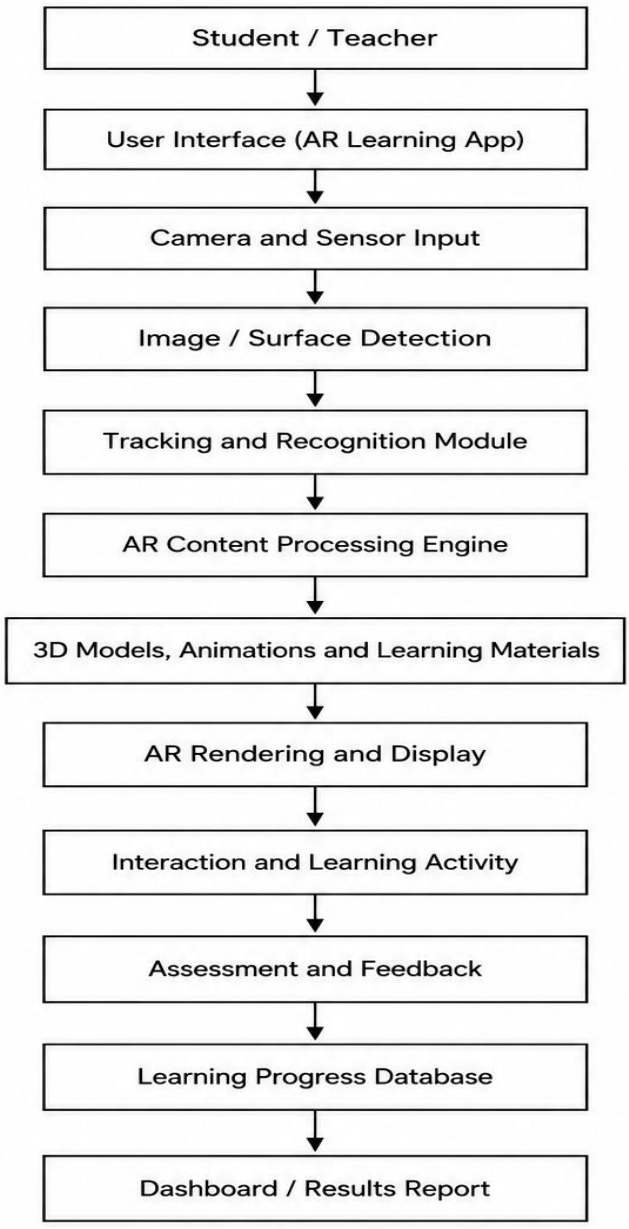


Fig. 3.2: System Architecture of Augmented Reality in Education

The main components of an Augmented Reality (AR) Education System include the User Interface, Camera and Sensors, Image Detection Module, Tracking and Recognition Module, AR Content Processing Engine, 3D Models and Animations, AR Rendering and Display Module, Assessment and Feedback Module, and Learning Progress Database. These components work together to combine virtual educational content with the real-

world environment, provide interactive learning experiences, assess student understanding, and display learning results.

1. Technologies and Techniques Used:

Augmented Reality in Education can be developed using technologies such as Augmented Reality (AR), Artificial Intelligence (AI), computer vision, 3D modelling, mobile applications, cameras, sensors, and cloud databases. The camera captures the real-world environment, while image detection and tracking techniques identify objects or surfaces. The AR processing engine loads educational content, such as 3D models, animations, and text. Rendering technology overlays the virtual content onto the real-world view. The final technology selection depends on the project requirements, available devices, and learning objectives.

2. Example of Augmented Reality in Education:

Consider a classroom where students are learning about the human body, the solar system, or the structure of a plant. Using an AR application on a smartphone or tablet, students can view interactive 3D models of these topics. For example, when studying the solar system, students can explore virtual planets, observe their positions, and rotate the models to understand their structure. This makes complex concepts easier to visualize and helps improve student engagement and understanding.

3. Interactive Learning and Student Assessment:

The AR education system displays virtual educational content through a mobile application or other compatible device. Students can interact with 3D models, animations, and virtual objects to explore different concepts. The application may provide quizzes, exercises, and feedback to assess student understanding. Learning progress and assessment results can be stored in a database and displayed on a dashboard, if these features are supported by the application. This process can make learning more engaging, improve visualization, and help students understand difficult concepts more effectively.

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages of AR in Education

1. Better Visualization

AR can display 3D objects and processes that are difficult to understand from flat diagrams alone. Learners can inspect an object from different angles and connect the representation with an explanation.

2. Increased Engagement and Motivation

Interactive content can encourage learners to explore, ask questions, and participate. Engagement is most useful when it leads to meaningful learning activity rather than simply adding entertainment.

3. Support for Experiential Learning

AR can create opportunities to learn by doing, such as assembling a model, exploring a virtual laboratory, or following an illustrated procedure.

4. Immediate and Contextual Information

Digital labels, hints, and instructions can appear at the point where they are needed. This can help learners connect information to real objects and environments.

5. Flexible Practice

Learners may repeat an activity and examine a model at their own pace. Some applications also provide prompts or feedback during practice.

6. Support for Different Subjects

AR is adaptable to science, mathematics, languages, history, geography, technical training, and other areas, provided the content is designed for the subject and learner level.

4.2 Disadvantages and Challenges

1. Cost and Device Availability

Compatible devices, software licenses, and custom content can require investment. Not all students have access to a suitable phone, tablet, or reliable internet connection.

2. Technical Problems

Tracking errors, device incompatibility, battery use, software bugs, and weak connectivity may interrupt a lesson.

3. Teacher Training

Teachers may need time and support to select appropriate applications, manage classroom activities, troubleshoot devices, and assess learning.

4. Cognitive Overload and Distraction

Too many labels, animations, sounds, or interactive features can overwhelm learners or divert attention from the lesson's main idea.

5. Accessibility and Inclusion

Some AR applications may not work well for learners with visual, hearing, motor, or other accessibility needs. Designers should provide alternatives and accessible content.

6. Privacy and Safety

AR apps may use camera, location, or account permissions. Institutions should review data collection, consent, age suitability, and safe use of devices.

7. Content Quality and Maintenance

Poorly designed models may contain errors or confuse learners. Applications also need updates and support as devices and operating systems change.

CONCLUSION

Augmented Reality offers a promising way to connect digital learning content with the physical environment. It can help students visualize complex structures, explore interactive models, practice procedures, and engage with information in context. Research reviews have reported potential benefits for engagement, motivation, spatial understanding, collaboration, and learning outcomes, especially when AR activities are carefully designed around educational objectives.

At the same time, AR introduces practical challenges. Costs, device availability, connectivity, technical usability, teacher training, accessibility, privacy, and the risk of cognitive overload can affect successful implementation. These issues should be considered before an institution adopts AR on a large scale.

In conclusion, AR should be treated as a supporting educational tool rather than a replacement for teachers, textbooks, laboratory work, or effective teaching practice. Its value depends on appropriate content, thoughtful lesson design, equitable access, and evaluation of student learning. When these conditions are addressed, AR can contribute to more interactive and meaningful educational experiences.

CHAPTER 5

PROJECT INITIATION

5.1 Description of Problem Statement

Many educational concepts are difficult to explain using only text, static pictures, or two-dimensional diagrams. Students may struggle to visualize three-dimensional structures, understand dynamic processes, or connect theoretical information with real objects. Practical demonstrations may also be limited by time, equipment, cost, safety, or access to laboratories and field sites.

Augmented Reality can provide interactive digital visualizations that appear within the learner's real environment. However, adoption is affected by the cost and availability of devices, technical issues, the quality of learning content, teacher readiness, accessibility, and the possibility of distraction. Therefore, there is a need to examine how AR can be applied effectively in education while identifying its benefits and implementation challenges.

5. 2 Scope and Objectives

a) Scope of the Study:

This report focuses on the educational uses of AR in schools, colleges, and training environments. It considers mobile and marker-based AR, 3D visualisation, interactive learning, and context-sensitive educational information. It reviews reported benefits and challenges using published literature. It does not claim to develop or experimentally test a new AR application; practical implementation would require a separate design, development, and evaluation phase.

b) Objectives:

1. To explain the meaning, types, and basic working principle of Augmented Reality.
2. To identify major applications of AR across educational subjects and settings.
3. To examine the reported benefits of AR for student engagement, visualization, and learning.
4. To identify technical, financial, pedagogical, accessibility, and privacy challenges.
5. To review selected research papers related to AR in education.
6. To suggest considerations for responsible and effective implementation of AR in educational institutions.

5.3 Stakeholders

Stakeholder	Role / Interest
Students	Use AR activities to explore course content and practice skills; provide feedback on usability.
Teachers / Faculty	Select learning goals, facilitate activities, support students, and assess outcomes.
Educational Institutions	Provide devices, connectivity, policies, technical support, and funding.
AR Developers / Content Designers	Create accurate, usable, accessible, and curriculum-aligned AR experiences.
Parents / Guardians	Support safe and age-appropriate use where applicable.
IT Support Staff	Install applications, manage devices, troubleshoot problems, and maintain security.
Researchers / Evaluators	Evaluate usability, engagement, learning outcomes, and implementation barriers.

5.4 Platform / Equipment / Resources Identification

1. Platform

An educational AR activity may run on Android or iOS smartphones and tablets. Depending on the project, AR content may be created using a suitable AR development platform or accessed through an existing educational application. A computer is useful for preparing content, managing files, and developing or testing applications.

2. Equipment

- Smartphone or tablet with a compatible camera and sensors.
- Computer or laptop for content preparation and development.
- Stable internet connection for downloading content or accessing online resources, where required.
- Printed markers or image targets if the selected application uses marker-based AR.
- Headphones or other classroom equipment if audio guidance is included.

3. Software and Resources

Resource	Example / Option	Purpose
Operating system	Windows, Android, or iOS	Run development tools or the AR learning application.
AR development tool	Unity with AR Foundation, or another suitable AR platform	Builds interactive AR scenes and experiences.
3D content tools	Blender or existing educational 3D assets	Creates or prepares 3D models.
Programming language	C# when using Unity; other languages depending on the platform	Implement interactions and application logic.
Learning materials	Textbooks, lesson plans, diagrams, verified 3D models	Keeps AR content aligned with curriculum and learning objectives.
Testing resources	Compatible devices and student/teacher feedback forms	Checks performance, usability, and educational value.
Human resources	Teacher, subject expert, developer, and evaluator	Ensures content accuracy, classroom fit, and effective assessment.

S.S.V.P.S's B. S. Deore Polytechnic, Dhule Department of Computer Engineering *Academic Year : 2026-2027*		
SEMINAR AND PROJECT INITIATION COURSE		
Sem : CO 5 K	**SPI – 315003**	**Date : 17/10/2026**
Topic : Augmented Reality in Education: Applications, Benefits and Challenges		

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Contents

- Abstract
1. Introduction
2. Literature Survey
3. Information to Seminar topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6. 1 Description of problem statement.
   - 6. 2 Scope and objectives.
   - 6. 3 Stake holder
   - 6.4 Seminar execution plan
   - 6.5 Platform/ Equipment/ Resources identification.
1. Bibliography
2. References

2

Abstract

- Augmented Reality (AR) adds 3D models, text and videos on top of the real world using a phone, tablet or AR glasses.
- In education, AR helps students see difficult topics (like the human heart or the solar system) instead of only reading about them.
- This seminar explains the applications, benefits and challenges of AR in education.
- Cost, teacher training and distraction are the main challenges for using AR in schools.

3

1. Introduction

- **What is AR?** Technology that shows digital objects on top of the real world in real time.
- **How does it work?** The phone camera and sensors find the place, and the app shows 3D content on the screen.
- **Example:** A student points a phone at a cell picture in the book and a 3D cell appears and can be rotated.
- **Why in education?** Many topics are hard to imagine from books and boards only.
- **Our topic:** Applications, benefits and challenges of AR in education.

4

2. Literature Survey

Sr.	Author and Year	Method used	Finding
1	Milgram and Kishino, 1994 [1]	Defined the reality–virtuality continuum for mixed reality displays	Shows where AR lies between real and virtual worlds
2	Azuma, 1997 [2]	Survey that defined AR by three features (real + virtual, live, 3D)	Gave the standard definition of AR
3	Billinghurst and Kato, 2002 [3]	Studied collaborative AR for face-to-face and remote work	AR helps groups work and learn together
4	Kaufmann and Schmalstieg, 2003 [4]	Construct3D: AR tool for geometry teaching with students	Students could see and build 3D shapes
5	Dunleavy et al., 2009 [5]	AR simulations for teaching and learning in classrooms	Engaging, but technical and classroom problems found
6	Wu et al., 2013 [6]	Review of AR roles, approaches and issues in education	Showed AR chances and challenges for teaching

5

2. Literature Survey

Sr.	Author and Year	Method used	Finding
7	Bacca et al., 2014 [7]	Systematic review of AR trends in education research	Motivation and learning gain are main benefits
8	Radu, 2014 [8]	Meta-review of AR learning studies with cross-media analysis	Better learning and interest, but attention and usability issues
9	Diegmann et al., 2015 [9]	Literature review of benefits of AR in education	Grouped benefits like motivation and better understanding
10	Akçayır and Akçayır, 2017 [10]	Systematic review of advantages and challenges of AR	Better achievement and attitude; usability and cost problems
11	Ibáñez and Delgado-Kloos, 2018 [11]	Systematic review of AR for STEM learning	Positive results; more teaching guidance needed
12	Garzón et al., 2019 [12]	Meta-analysis of AR in educational settings	AR has a medium positive effect on learning

6

3. Information to seminar topic

What is Augmented Reality?

1. AR adds digital things (3D models, text, video) to the real world.
2. It works live on a phone, tablet or AR glasses.
3. The student still sees the real classroom.
4. VR replaces the real world, but AR only adds to it.

Types of AR

- Marker-based – camera scans an image or QR code
- Markerless – uses GPS and sensors, no marker
- Location-based – shows content at a place, e.g. field trips

7

3.1 Applications of AR in education

AR is used in many subjects:

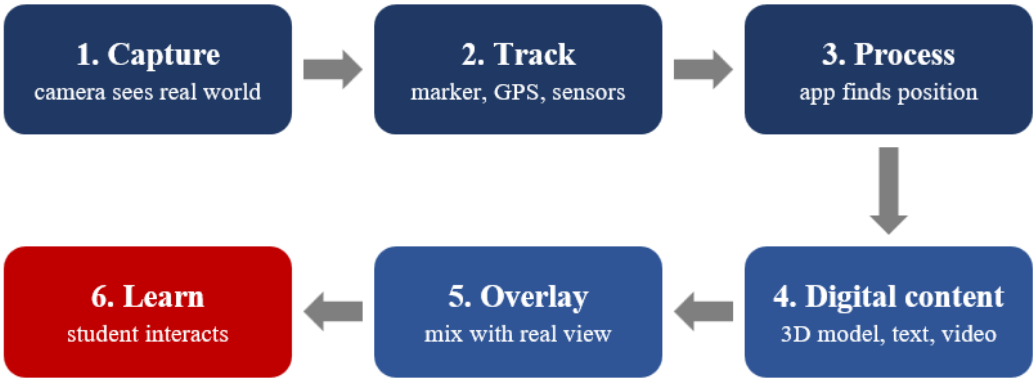
- **Science** – see 3D models of cells, organs and planets.
- **Mathematics** – see geometry shapes in 3D and rotate them.
- **Medical and engineering** – practise on virtual body parts and machines.
- **History and geography** – bring monuments and maps to life.
- **Language and kids** – AR flashcards and story books.

Students learn by seeing and interacting, not only by reading.

8

3.2 How AR works in a classroom

Steps of AR



Example: point the phone at a book picture and a 3D heart appears that you can rotate.

9

3.3 AR tools and apps used

Devices

- Smartphone or tablet camera is most common. AR headsets and smart glasses are costly and less common in schools.

Software and platforms

- Google ARCore (Android), Apple ARKit (iPhone), Unity with AR Foundation and Vuforia are used to build AR apps.

Education apps

- Apps like Merge Cube and Complete Anatomy show 3D models in real space.

10

4. Advantages / Disadvantages

Advantages

- Makes lessons interesting.
- Explains difficult topics.
- Increases student motivation.
- Safe practice (labs, surgery).
- Can be used anywhere with a phone.

Disadvantages

- Devices and apps are costly.
- Teachers need training.
- Can distract students.
- Needs good phones and internet.
- Less content for local syllabus.

11

5. Conclusion

- AR adds digital content to the real world and makes learning interactive.
- It is used in science, maths, medicine, history and more.
- Main benefits: better understanding, interest and motivation.
- Main challenges: cost, teacher training and distraction.
- Future: AR with AI and cheaper devices can reach more schools.

12

6. Project Initiation

- **Project title:** Augmented Reality in Education: Applications, Benefits and Challenges
- **Subject:** Seminar and Project Initiation (SPI – 315003)
- **Group members:** Deepak Bhikaji More, Manjeet Haresh Kundal, Bhumi Rajesh Mahajan
- **Next slides:** problem, scope, objectives, stakeholders, plan and resources.

13

6.1 Description of problem statement

Problem

- Many topics (anatomy, space, chemistry) are hard to understand from books only.
- Schools cannot always give labs, models or field trips.
- Students lose interest in plain lectures.

Problem statement

To study how Augmented Reality can be used in education, with its applications, benefits and challenges.

14

6.2 Scope and objectives

Objectives

- To understand what AR is and how it works.
- To study AR applications in different subjects.
- To find benefits of AR for students and teachers.
- To list challenges and possible solutions.

Scope

- Covers AR use in schools, polytechnics and colleges.
- Can be extended to a simple AR demo app later.

15

6.3 Stake holder

Stakeholder	How they are related
Students (learners)	Learn using AR lessons and apps
Teachers / institutes	Use AR in class; need training
AR app developers	Make AR content and education apps
Project guide / teacher	Guides and checks our seminar
HOD / Department	Gives approval

16

6.4 Seminar execution plan

Step	Work to do	Time
1	Select topic and study AR basics	Week 1
2	Literature survey (read papers)	Week 2
3	Study AR applications in education	Week 3
4	Collect benefits and challenges	Week 4
5	Study AR tools (ARCore, ARKit, Unity)	Week 5
6	Try a simple AR app / demo	Week 6
7	Prepare content and diagrams	Week 7
8	Make the presentation slides	Week 8
9	Practice and corrections	Week 9
10	Make report and final presentation	Week 10

17

6.5 Platform / Equipment / Resources

Hardware

- Laptop / computer with internet and a smartphone with camera

Software

- Unity with AR Foundation (for a simple demo)
- Google ARCore / Apple ARKit supported phone

Study material

- Research papers (Google Scholar, IEEE Xplore)
- Free 3D models (Sketchfab)

18

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20

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Thank you

CHAPTER 6

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CHAPER 7

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Name of student
1	24610960217	Bhumi Rajesh Mahajan
2	24610960222	Deepak Bhikaji More
3	24610960215	Manjeet Haresh Kundal

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number		
			388525	388530	388523
1	Selection Topic / Theme of seminar	5			
2	Literature review and data presentation	5			
3	Quality of Preparation and innovativeness	5			
4	Q-A handling	5			
5	Time Management	5			
6	Seminar Presentation report	10			
7	Selection of Theme of Problem Statement and its innovativeness	5			
8	Stages of development of Action plan	5			
9	Prototyping	5			
	Total	**50**			
	Scaled to 25				

Sign :

Name : ----------

(Course Expert/s)